Short Question Assignment Solutions

Question 1: Matrix Dimensions in SVD

Full SVD shapes: U is 1000×1000, Σ is 1000×50, V^T is 50×50

Question 2: Truncated SVD and Data Compression.

A) The truncated Σ_k matrix is 5×5.

B) We keep the largest singular values because they correspond to the directions of maximum variance in the dataset. They represent the strongest patterns (the signal). While the smallest singular values represent tiny, random fluctuations (the noise)

Question 3: SVD vs. Eigendecomposition

For a symmetric, positive semi-definite matrix, SVD and Eigendecomposition are mathematically indentical. Therefore, $Q = U = V$, and the eigenvalues perfectly equal the singular values ($\Lambda = \Sigma$).

Question 4: PCA under the Hood

The matrix V^T (the right singular vectors) directly contains the Principal Components. if rows represent the axes of maximum Variance

Question 5: The condition Number and Collinearity.

if two features are Perfectly Correlated, the matrix loses full rank, and the smallest singular value will exactly $\sigma_{min} = 0$. The Condition Number ($\sigma_{max}/0$) approches infinity. This tells you the system is Perfectly singular, and standard Linear Regression will fail (crash due to an inability to invert the matrix) unless regularization technique like Ridge/Lasso are applied.